Feature Flag: Betting Deposit Age Gate

Banking application: hiding betting-platform deposits from users under 18

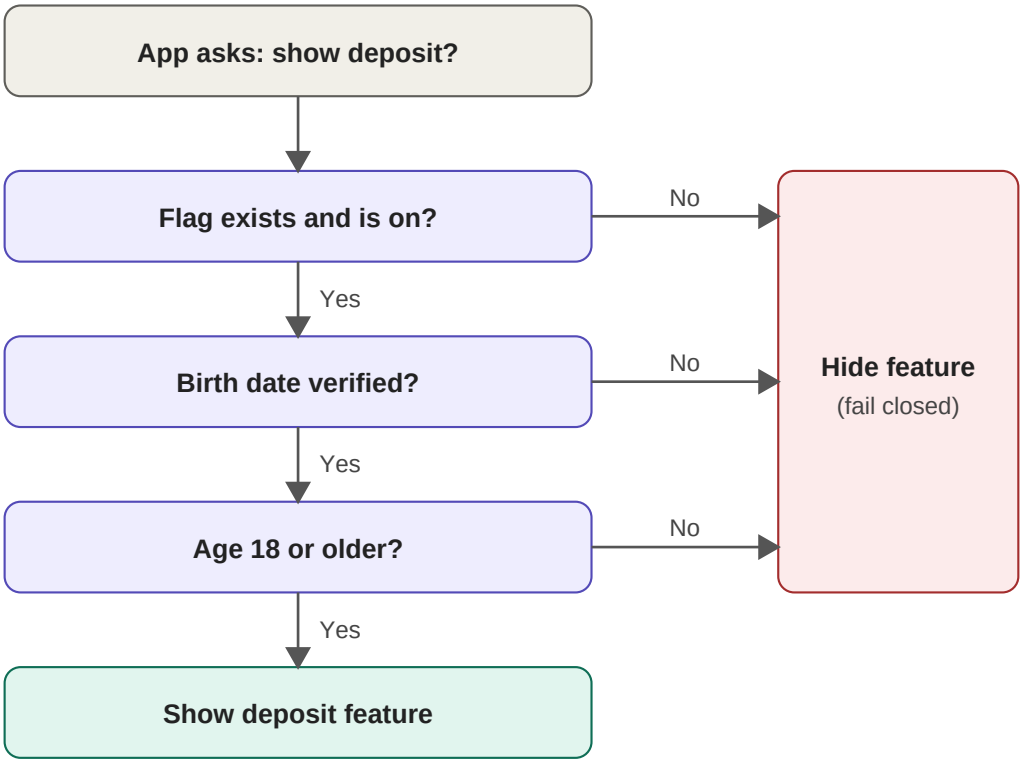
A. Creating the feature (admin steps)

1. Open the admin dashboard > Feature Management > "Create feature".
2. Enter the key `betting_deposit`.
3. Add a targeting rule. For this feature: `age >= 18`.
4. Set the default behaviour to "off" for anyone who doesn't match the rules.
5. Choose the rollout: everyone who matches, or a percentage or segment for a gradual release.
6. Save as Draft, then test with a few internal accounts (a 17-year-old and an 18-year-old test user).

B. How the system decides access (step by step)

1. **Request arrives.** The app sends the user's ID and the flag key (`betting_deposit`) to the flag service, for example when the home screen loads.
2. **Load the flag.** The service fetches the active version of the flag, usually from a cache that refreshes within seconds of an admin change. If the flag doesn't exist, the answer is no.
3. **Check the master switch.** If `enabled = false`, stop and return no. This is the kill switch.
4. **Build the user context.** The service gathers the facts the rules need from trusted sources: the date of birth stored on the user's record, country, account status.
5. **Calculate age.** Age is computed from the date of birth against today's date in a fixed timezone (e.g. WAT), not stored as a number that goes stale.
6. **Evaluate the rules.** Each rule is checked against the user context (`age >= 18`). All must pass. If any attribute is missing, that rule fails.
7. **Apply rollout.** If the flag is released gradually, hash the user ID into a bucket from 0 to 99 and compare it to the rollout percentage. The same user always lands in the same bucket, so they don't flip in and out of the feature. Safety rules like age are checked before this step, so rollout can never let a minor in.
8. **Return the result and log it.** The service returns allowed or denied with a reason code (such as `UNDER_AGE` or `DOB_NOT_VERIFIED`), and the decision is recorded for audit.
9. **Enforce it twice.** The app hides or shows the deposit button based on the result, and the deposit API calls the same service again before processing any money. This stops a minor from calling the endpoint directly.

C. Flowchart



D. Problems the system needs to handle

- **Missing or unverified date of birth.** Hide the feature (deny by default). Don't assume the user is an adult.
- **Birthday edge cases.** Someone turning 18 tomorrow is still 17 today. Calculate age against the current date in one agreed timezone (e.g. WAT) rather than just subtracting years.
- **Users who turn 18.** Evaluate at request time instead of storing a cached "allowed" result, so access appears on the birthday without manual action.
- **Wrong or tampered date of birth.** Take it from the user's stored record, never from a field the user can edit, and lock it once recorded.
- **Hiding is not enough.** Enforce the rule in the backend deposit endpoint too, because a minor could call the API directly. The UI flag is only a convenience.
- **Stale caches.** A kill switch flipped by an admin must reach every server and app session quickly.
- **Joint, child, or guardian accounts.** Decide whose age counts, and likely block the feature on all minor-linked accounts.
- **Audit and compliance.** Log every decision for sensitive flags (user, rule, result) so you can show regulators the control worked. Also check local gambling regulations, which may set rules beyond age.